In [1]:

```
#Loads scripts, libraries, etc.
print('Hello, lets get started')
%matplotlib widget
import importlib
import PyMca5
import os
from analyze_sgm_bsky_data import analyze_sgm_bsky_data
from alignment_utils import interactive_roll_align
from sum_mcc1_spectra import sum_mcc1_spectra
from save_pymca_stack_h5 import save_pymca_stack_h5
from plot_sgm_bsky_data import plot_sgm_bsky_data
from pca_xanes_analysis import pca_xanes_analysis
from interactive_cluster_merger import interactive_cluster_merger
from pca_xanes_analysis import run_pca_all_detectors
from cluster_xanes_analysis import run_clustering_all_detectors
from save_pymca_4d_stack_h5 import save_pymca_4d_stack_h5
from pca_stack_analysis import pca_stack_analysis
from cluster_pca_spectra import cluster_pca_spectra
from pca_xanes_analysis import pca_xanes_analysis
from cluster_xanes_analysis import cluster_xanes_analysis
os.environ["KMP_DUPLICATE_LIB_OK"] = "True"
os.environ["OMP_NUM_THREADS"] = "2"

print('Done loading libraries')
```

```
Hello, lets get started

[VERSION] Dashboard Engine v2.5 (ROI FIXED)

Done loading libraries
```

In [2]:

```
# Loads either a stack or map, including survey maps for analysis
# Browse to the file
# Summary of the stack shown in console
sgm_data = analyze_sgm_bsky_data()
```

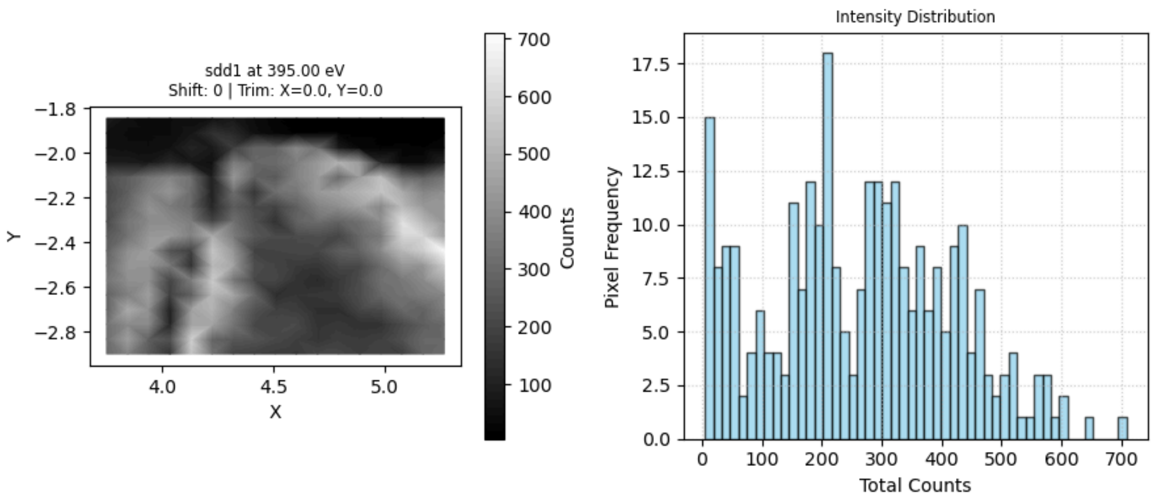
```
Analyzing File: D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1
s_stack\2026-04-24_223037_stack_data.h5

--- Scan Analysis Summary ---
Energies (182 points): [395.  395.5 396.  396.5 397.  397.5 398.5 398.6 398.7 398.8
398.9 399.
 399.1 399.2 399.3 399.4 399.5 399.6 399.7 399.8 399.9 400.  400.1 400.2
 400.3 400.4 400.5 400.6 400.7 400.8 400.9 401.  401.1 401.2 401.3 401.4
 401.5 401.6 401.7 401.8 401.9 402.  402.1 402.2 402.3 402.4 402.5 402.6
 402.7 402.8 402.9 403.  403.1 403.2 403.3 403.4 403.5 403.6 403.7 403.8
 403.9 404.  404.1 404.2 404.3 404.4 404.5 404.6 404.7 404.8 404.9 405.
 405.1 405.2 405.3 405.4 405.5 405.6 405.7 405.8 405.9 406.  406.1 406.2
 406.3 406.4 406.5 406.6 406.7 406.8 406.9 407.  407.1 407.2 407.3 407.4
 407.5 407.6 407.7 407.8 407.9 408.  408.1 408.2 408.3 408.4 408.5 408.6
 408.7 408.8 408.9 409.  409.1 409.2 409.3 409.4 409.5 409.6 409.7 409.8
 409.9 410.  410.1 410.2 410.3 410.4 410.5 410.6 410.7 410.8 410.9 411.
 411.1 411.2 411.3 411.4 411.5 411.6 411.7 411.8 411.9 412.  412.1 412.2
 412.3 412.4 412.5 412.6 412.7 412.8 412.9 413.  413.1 413.2 413.3 413.4
 413.5 413.6 413.7 413.8 413.9 414.  414.1 414.2 414.3 414.4 414.5 414.6
 414.7 414.8 414.9 415.  415.5 416.  416.5 417.  417.5 418.  418.5 419.
 419.5 420. ]
Number of Images:       182
Energy Regions:         395.00-397.50: 0.5 eV, 398.50-415.00: 0.1 eV, 415.50-420.00:
0.5 eV
X array (shape):        (289,) (Nx: 17)
Y array (shape):        (289,) (Ny: 17)
Grid Dimensions:        17 x 17
----------------------------
Date:                   2026-04-24
Scan Name:              Au13_0013_n1s
Project:                40G14650
Beamline:               N/A
Polarization:           N/A
Grating:                N/A
Harmonic:               N/A
Strip:                  N/A
Command:                N/A
Coordinates:            X: 3.75 to 5.27, Y: -2.90 to -1.84
Exit Slit Gap:          N/A

MCC Files Found:        182

SDD Files Found:
  Detector sdd1: 182 files
  Detector sdd2: 182 files
  Detector sdd3: 182 files
  Detector sdd4: 182 files
-----------------------------------
```

In [3]:

```python
# Allows for the alignment of the stack to be checked
# Allows for removing some pixels ( as mm units) on the X and Y edges, which makes
# The new dimensions will be used in the plotting and PCA/CA
# This step can be skipped
interactive_roll_align(sgm_data, (30,50), use_color=False)
```

```
VBox(children=(HBox(children=(Dropdown(description='Detector:', options=('sdd1', 'sdd2', 'sdd3', 'sdd4'), valu…
```

align_fig_2725644874048




In [4]:

```
# Maps the fluorescence and sample TEY signal.  Can select which image for the map.
# Plots the fluorescence and TEY spectra from a stack- saves Raw and Normalized XAN
# Saves XRF spectra for each SDD as a csv file. Saves images (i.e., maps) for each
# Can change the ROI from the XRF spectrum to select the edge of interest. Each bin
# Can change the area selected on the image by dragging a rectangle or making a pol
# Note that the area selected will be where the spectra are derived from and saves
# To see the changes in the spectra need to click on the "Refresh Plots" button.
# Can use the internal I0 from the Au mesh (mcc1) or load an external I0 from a csv
# MCC1 is the I0 from Au mesh before the KB mirrors, MCC2 is from a photodiode behi
# which could be used as and Io, , MCC3 auxillary, MCC4 is TEY signal from the samp
# Can either show lines with or withput individual data points
# Can zoom in on spectra using tools on left.
# Mouse over an image or plot and get the X and Y coordinates.
# Automatically adjusts the contrast for each image.
# Plot as a scatter map (usual way) or as a heatmap.
# The default ROI on the XRF spectra is 0-255.  Drag left and right edges on the XR
# Each bin is about 10 eV.
# Calibrate the energy by adding an energy shift value - energy_shift = 0 as defaul
# To add detailed information to the header change use_full_metadata = True.
# Can display spectra with the data points or not- show_markers = False spectra plo
# Can change the display of the images, selecting grey scale or color

plot_sgm_bsky_data(sgm_data, as_scatter_plot=True, fixed_roi=False, mcc_to_map=[4],
#plot_sgm_bsky_data(sgm_data, as_scatter_plot=True, use_color=False, fixed_roi=Fals
```

```
  [Auto Select] Defaulting to middle energy: 407.00 eV
--- SDD Stack Dashboard: Au13_0013_n1s ---
  [Metadata] Full Headers: Disabled
  Location: D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack
  [Alignment] Roll Shift: 0 | Trim: X=0.000 mm, Y=0.000 mm
  Pass 1: Pre-loading energy dependence...
HBox(children=(Label(value='[ROI ENGINE v2.6 Ready] Dashboard Active.'), Button(button_style='success', descri…
VBox(children=(Canvas(layout=Layout(min_width='1450px'), toolbar=Toolbar(toolitems=[('Home', 'Reset original v…
```

```
VBox(children=(HBox(children=(IntSlider(value=91, description='Select Image Energy f
or Display:', layout=Layou…
VBox(children=(Canvas(layout=Layout(min_width='1450px'), toolbar=Toolbar(toolitems=
[('Home', 'Reset original v…
```

Out[4]:
```
<plot_sgm_bsky_data.Synchronizer at 0x27ad1f91550>
```

In [5]:
```python
# Saving for PCA/CA and for pyMCA analysis
# Need to select the appropriate roi for the edge of interest
print("PyMca version", PyMca5.__version__)
saved_pymca_stack=save_pymca_stack_h5(sgm_data, channel_roi=(37,55))
```

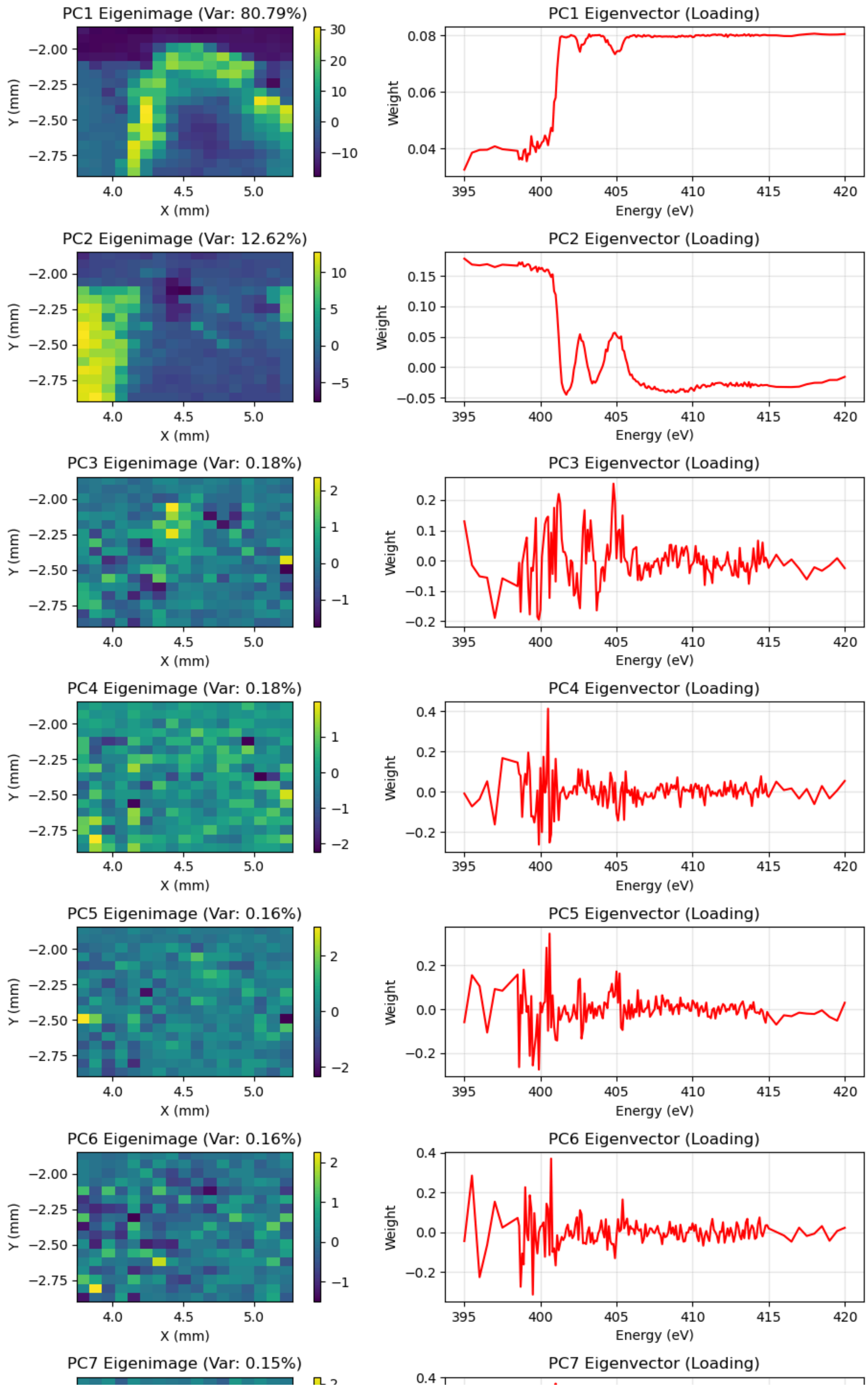
```
PyMca version 5.9.5
    -> No map_roi provided; using full scan range.
    -> Target Export ROI: X=[3.752, 5.268], Y=[-2.900, -1.845]
    -> Using user-selected I0 for normalization: Internal: mcc1
Processing 182 energy steps for 4 detectors...
    -> Progress: 10/182 energies processed.
    -> Progress: 20/182 energies processed.
    -> Progress: 30/182 energies processed.
    -> Progress: 40/182 energies processed.
    -> Progress: 50/182 energies processed.
    -> Progress: 60/182 energies processed.
    -> Progress: 70/182 energies processed.
    -> Progress: 80/182 energies processed.
    -> Progress: 90/182 energies processed.
    -> Progress: 100/182 energies processed.
    -> Progress: 110/182 energies processed.
    -> Progress: 120/182 energies processed.
    -> Progress: 130/182 energies processed.
    -> Progress: 140/182 energies processed.
    -> Progress: 150/182 energies processed.
    -> Progress: 160/182 energies processed.
    -> Progress: 170/182 energies processed.
    -> Progress: 180/182 energies processed.
    -> Progress: 182/182 energies processed.

Successfully saved PyMca-compatible HDF5 stack (multi-detector) to: D:/Data/Testing_
jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_
data_PyMca.h5
Datasets: sdd1, sdd2, sdd3, sdd4, sum, average
Default Signal: average
```

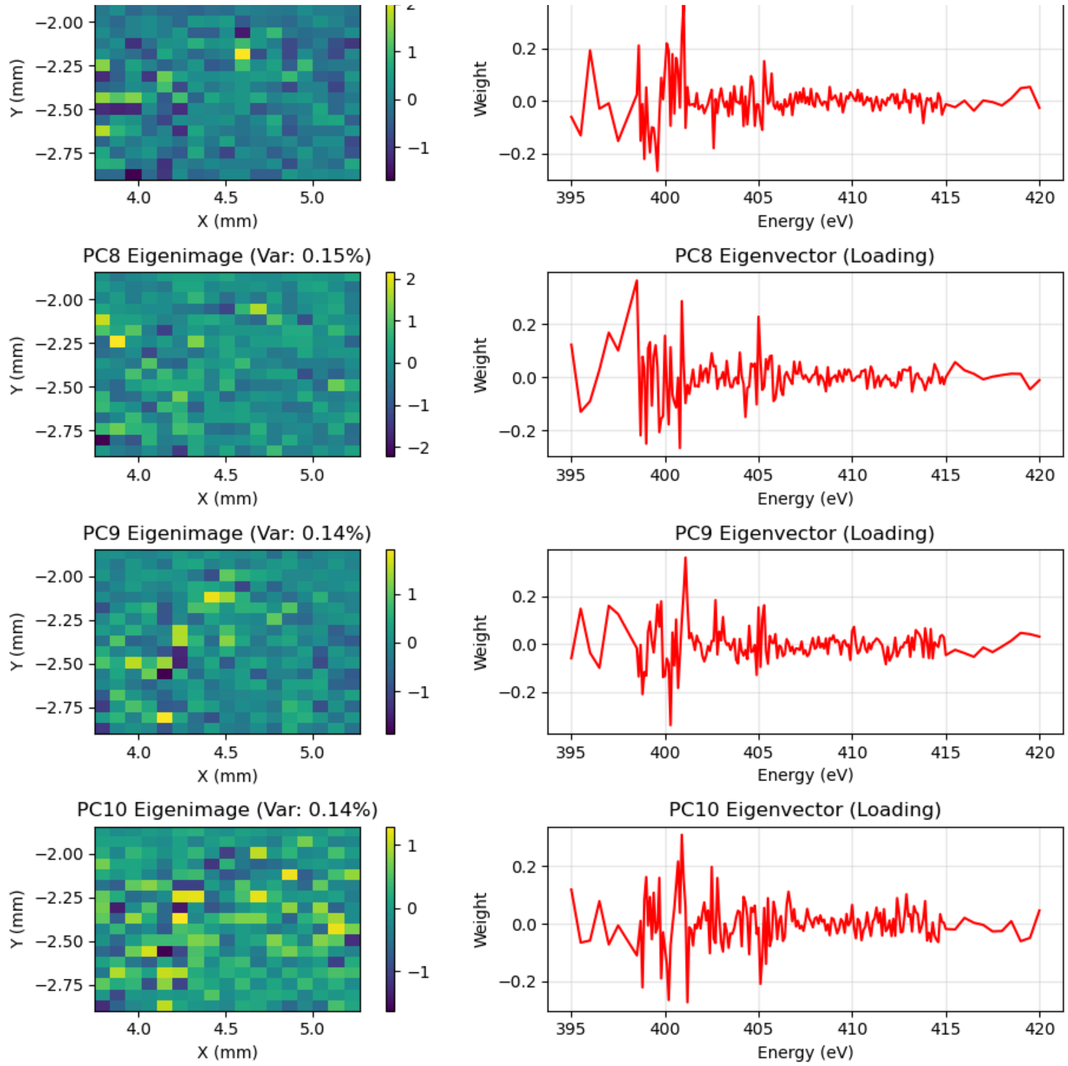
In [6]:
```python
# For single detector
# Determined the principal components for each SDD, or for the sum or average of ea
# Select the SDD and number of components.
# Displays the eigenimages and vectors

pca_results=pca_xanes_analysis(saved_pymca_stack, dataset_name ='sdd3', n_component
```

```
Loading dataset 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd3 stack shape: (17, 17, 182)
    -> sdd3 results saved to HDF5.
    -> Preview plot saved to: D:\Data\Testing_jupyter-python-axis2000\N_stacks\22302
9_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_sdd3_pca_preview.png
```

Figure

PCA Results for Stack: sdd3






In [7]:

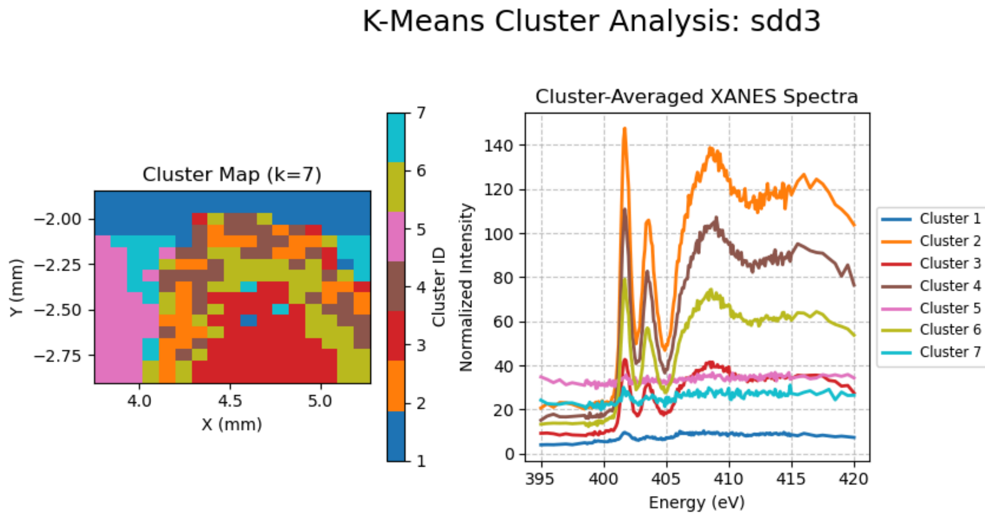
```
# For single detector
# Plots cluster maps and spectra for each detector and number of clusters selected
cluster_results=cluster_xanes_analysis(pca_results, dataset_name='sdd3', use_full_m
```

```
Loading PCA results for 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\threadpo
olctl.py:1226: RuntimeWarning:
Found Intel OpenMP ('libiomp') and LLVM OpenMP ('libomp') loaded at
the same time. Both libraries are known to be incompatible and this
can cause random crashes or deadlocks on Linux when loaded in the
same Python program.
Using threadpoolctl may cause crashes or deadlocks. For more
information and possible workarounds, please see
    https://github.com/joblib/threadpoolctl/blob/master/multiple_openmp.md

  warnings.warn(msg, RuntimeWarning)
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
```

```
    -> sdd3 cluster results saved.
    -> Preview plot saved to: D:/Data/Testing_jupyter-python-axis2000/N_stacks/22302
9_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_sdd3_cluster_preview.png
```

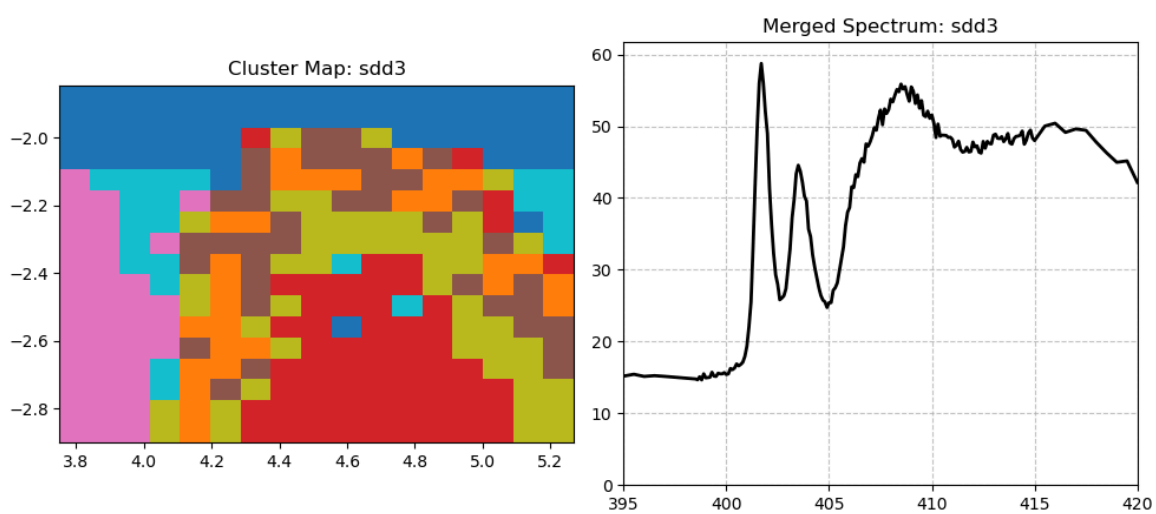
Figure




In [8]:
```python
# Allows for pixels to be exclude from the spectra
interactive_cluster_merger(cluster_results, dataset_name ='sdd3')
```

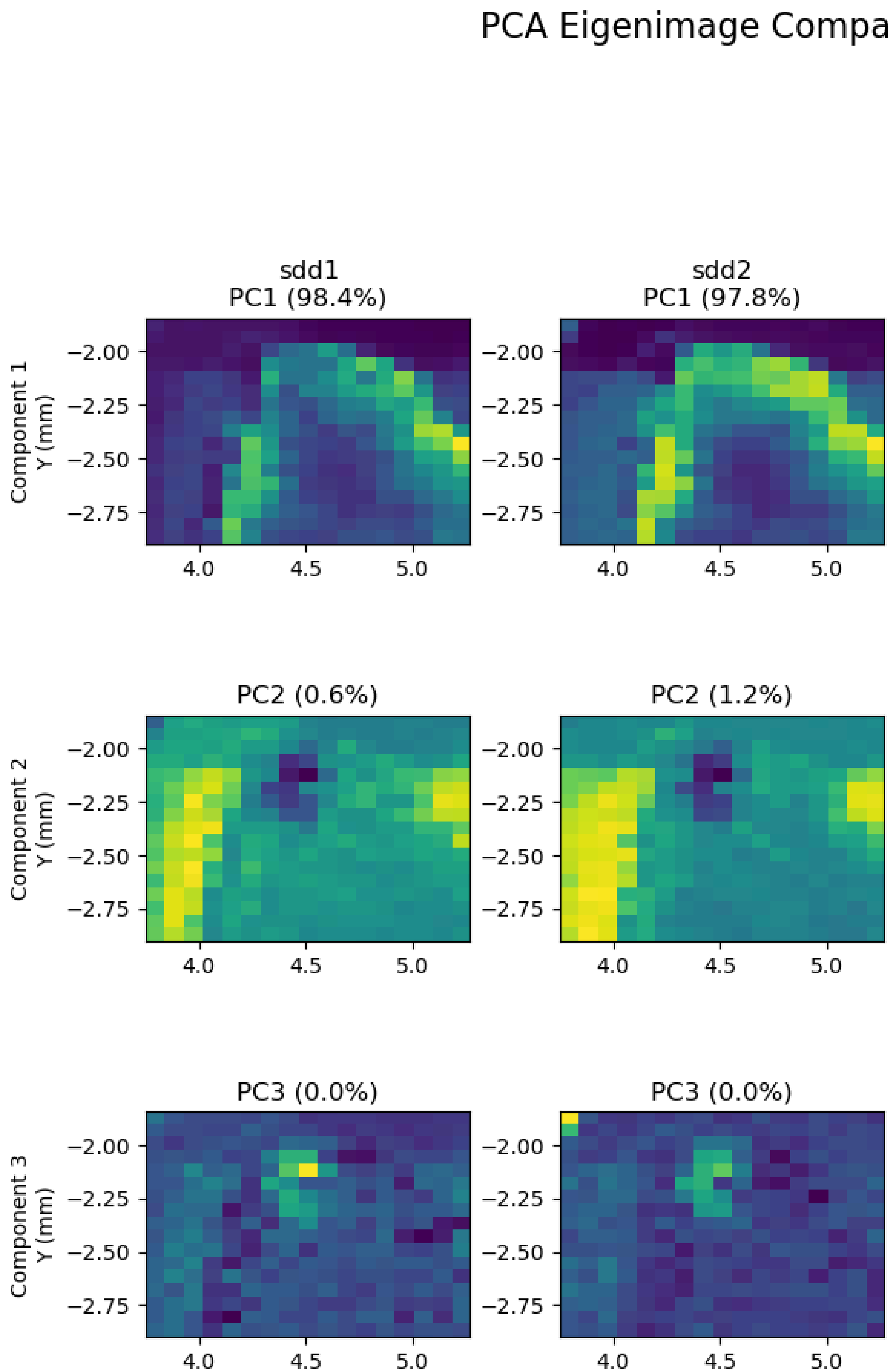
VBox()

merger_fig_2726597772912

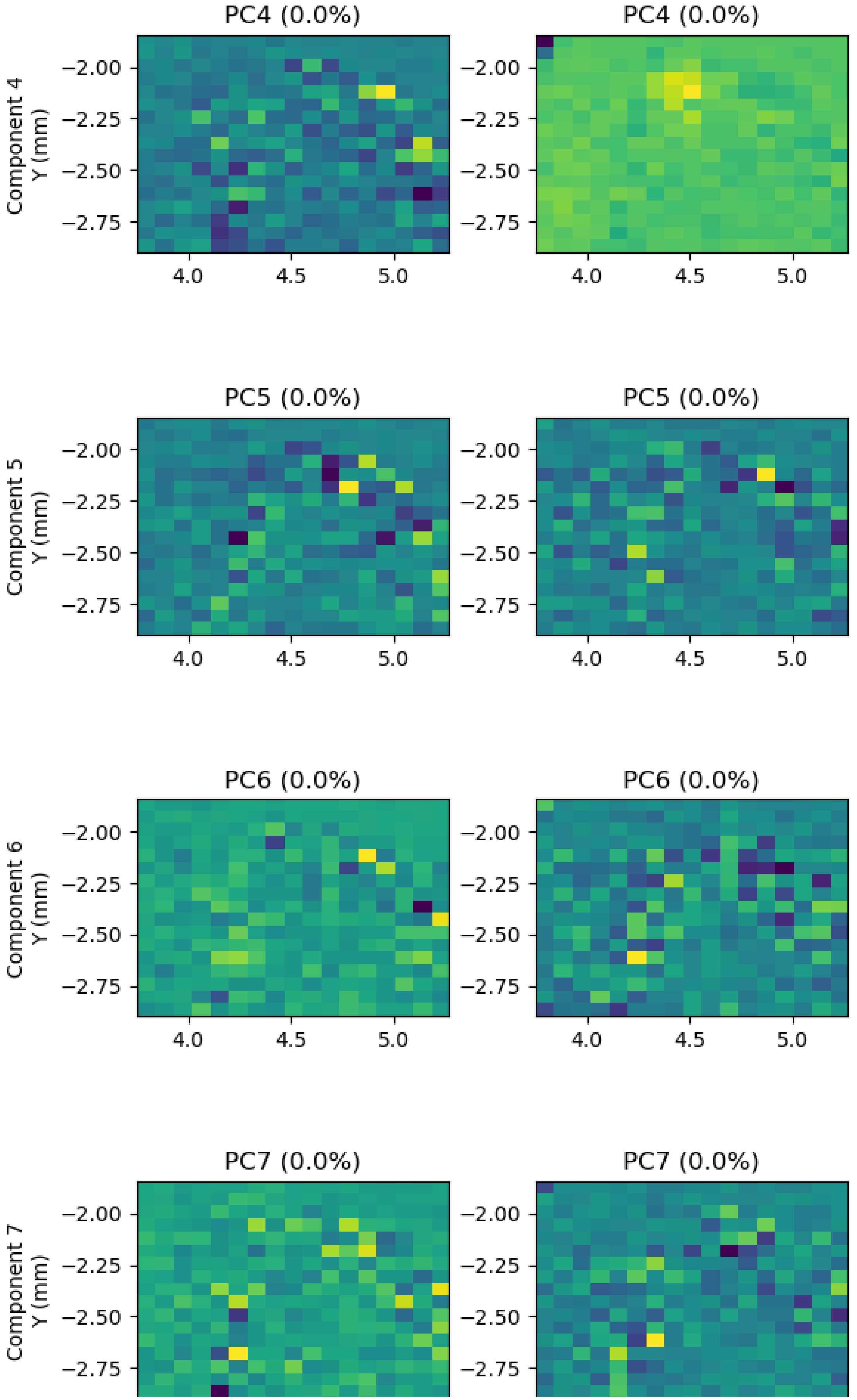


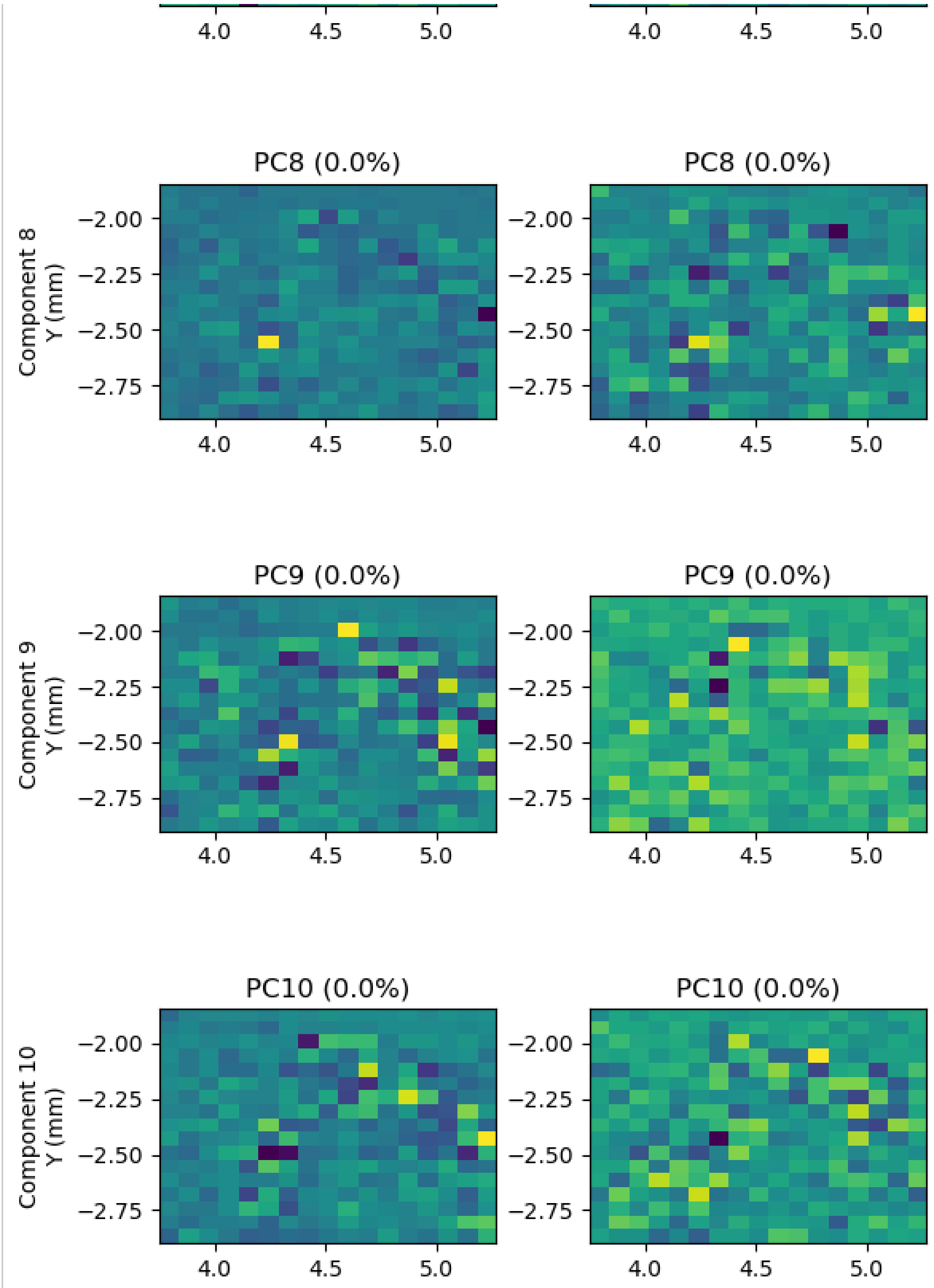

In [9]:
```python
# For all detectors
pca_results=run_pca_all_detectors(saved_pymca_stack, n_components=10)
```

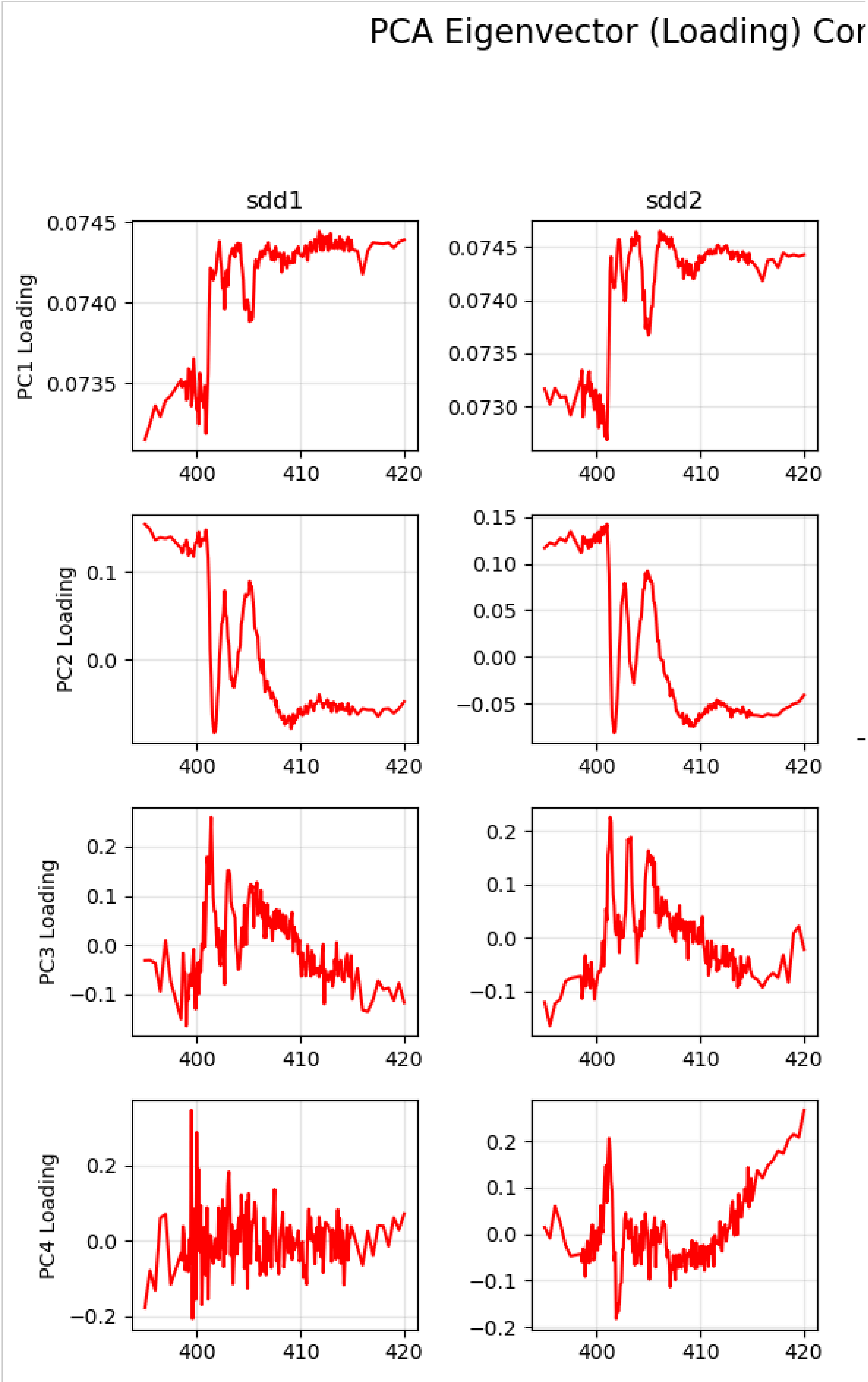
```
============================================================
Running Multi-Detector PCA Analysis
============================================================
Loading dataset 'sdd1' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd1 stack shape: (17, 17, 182)
    -> sdd1 results saved to HDF5.
Loading dataset 'sdd2' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd2 stack shape: (17, 17, 182)
    -> sdd2 results saved to HDF5.
Loading dataset 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd3 stack shape: (17, 17, 182)
    -> sdd3 results saved to HDF5.
Loading dataset 'sdd4' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd4 stack shape: (17, 17, 182)
    -> sdd4 results saved to HDF5.
Loading dataset 'average' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/2230
29_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> average stack shape: (17, 17, 182)
    -> average results saved to HDF5.

Multi-detector plots saved to:
  -> D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\202
6-04-24_223037_stack_data_PyMca_pca_comparison_images.png
  -> D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\202
6-04-24_223037_stack_data_PyMca_pca_comparison_vectors.png
```



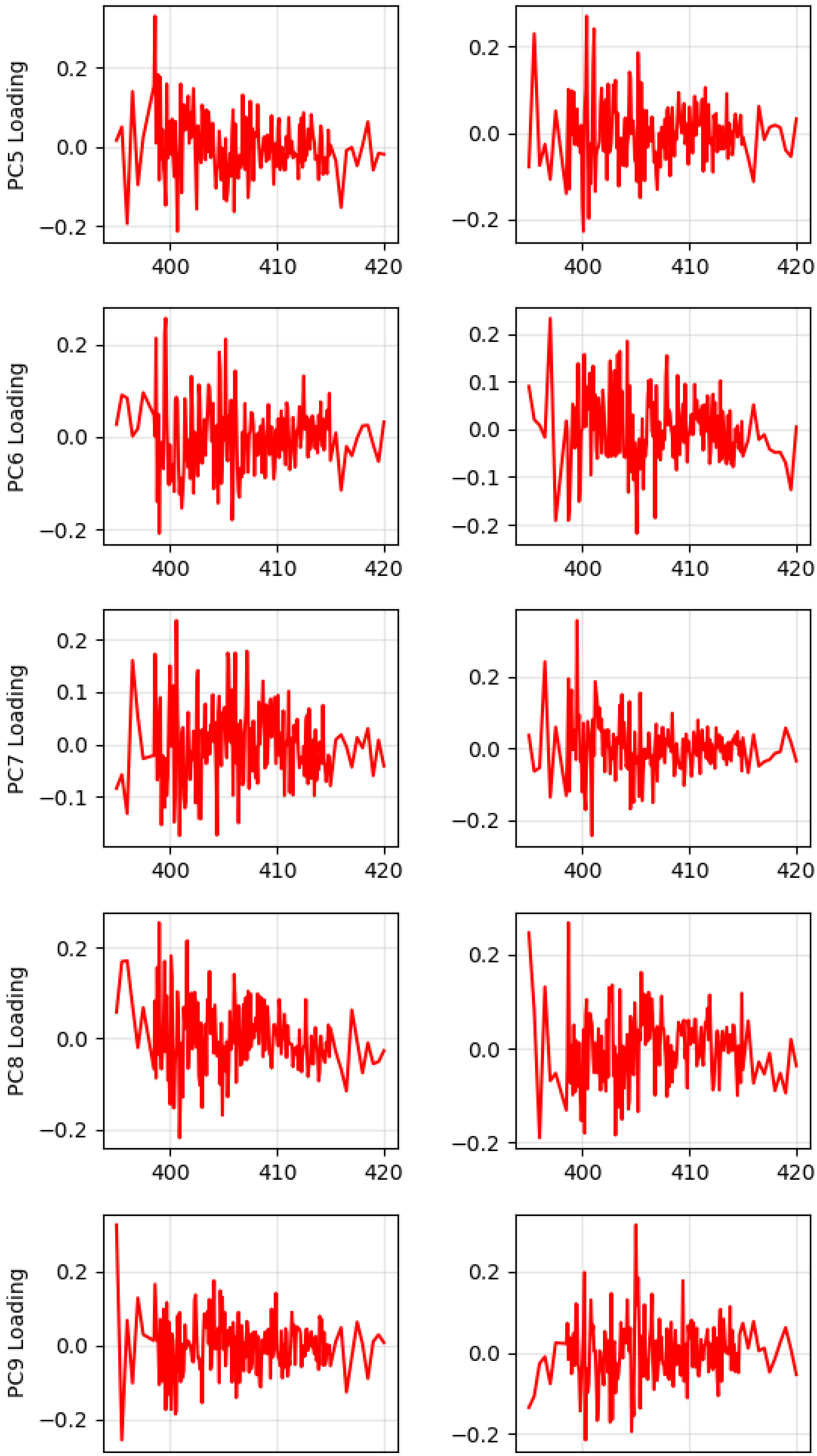
PCA Eigenimage Compar
sdd1
PC1 (98.4%)
sdd2
PC1 (97.8%)
PC2 (0.6%)
PC2 (1.2%)
PC3 (0.0%)
PC3 (0.0%)
Component 1
Component 2
Component 3
Y (mm)
−2.00
−2.25
−2.50
−2.75
4.0
4.5
5.0

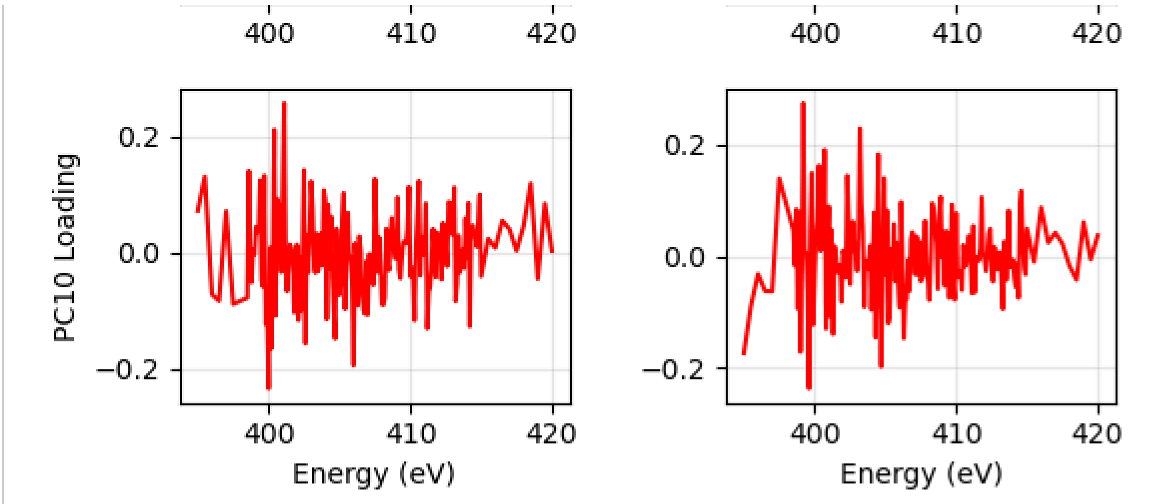

PC4 (0.0%)
PC4 (0.0%)
Component 4
Y (mm)
PC5 (0.0%)
PC5 (0.0%)
Component 5
Y (mm)
PC6 (0.0%)
PC6 (0.0%)
Component 6
Y (mm)
PC7 (0.0%)
PC7 (0.0%)
Component 7
Y (mm)
−2.00
−2.25
−2.50
−2.75
4.0
4.5
5.0


PC8 (0.0%)
PC8 (0.0%)
Component 8
Y (mm)
PC9 (0.0%)
PC9 (0.0%)
Component 9
Y (mm)
PC10 (0.0%)
PC10 (0.0%)
Component 10
Y (mm)
−2.00
−2.25
−2.50
−2.75
4.0
4.5
5.0



PCA Eigenvector (Loading) Cor
sdd1
sdd2
PC1 Loading
PC2 Loading
PC3 Loading
PC4 Loading
0.0745
0.0740
0.0735
0.0730
0.15
0.10
0.05
0.00
−0.05
0.1
0.0
0.2
−0.1
−0.2
400
410
420

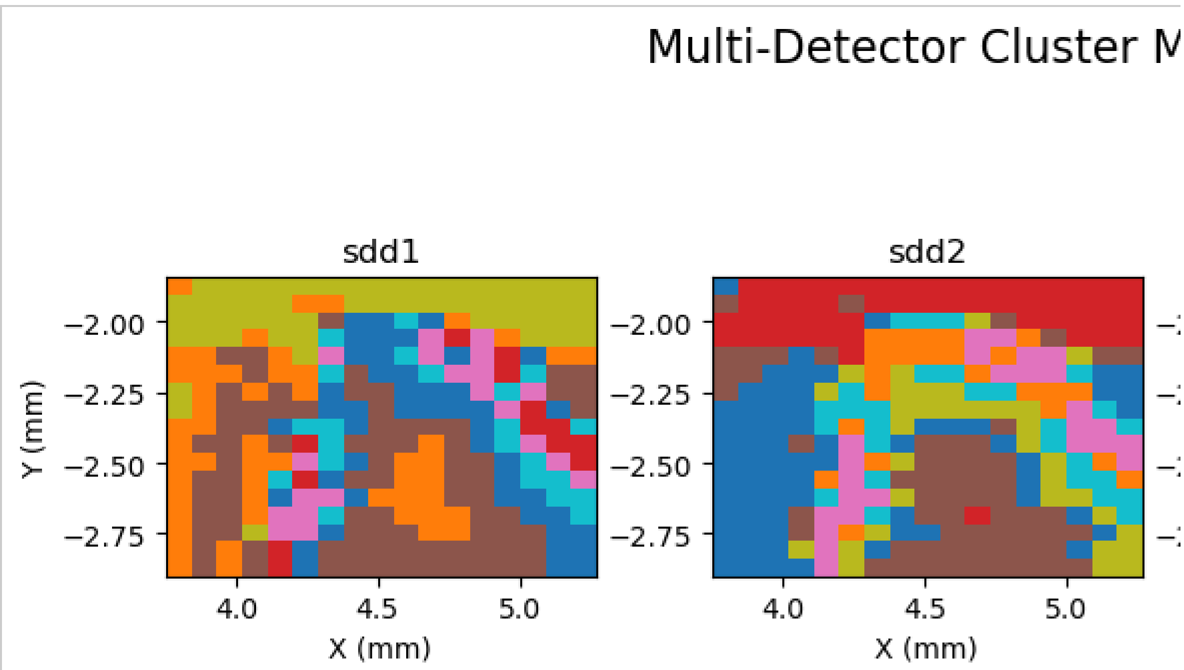

PC5 Loading
PC6 Loading
PC7 Loading
PC8 Loading
PC9 Loading
400
410
420
0.2
0.1
0.0
−0.1
−0.2

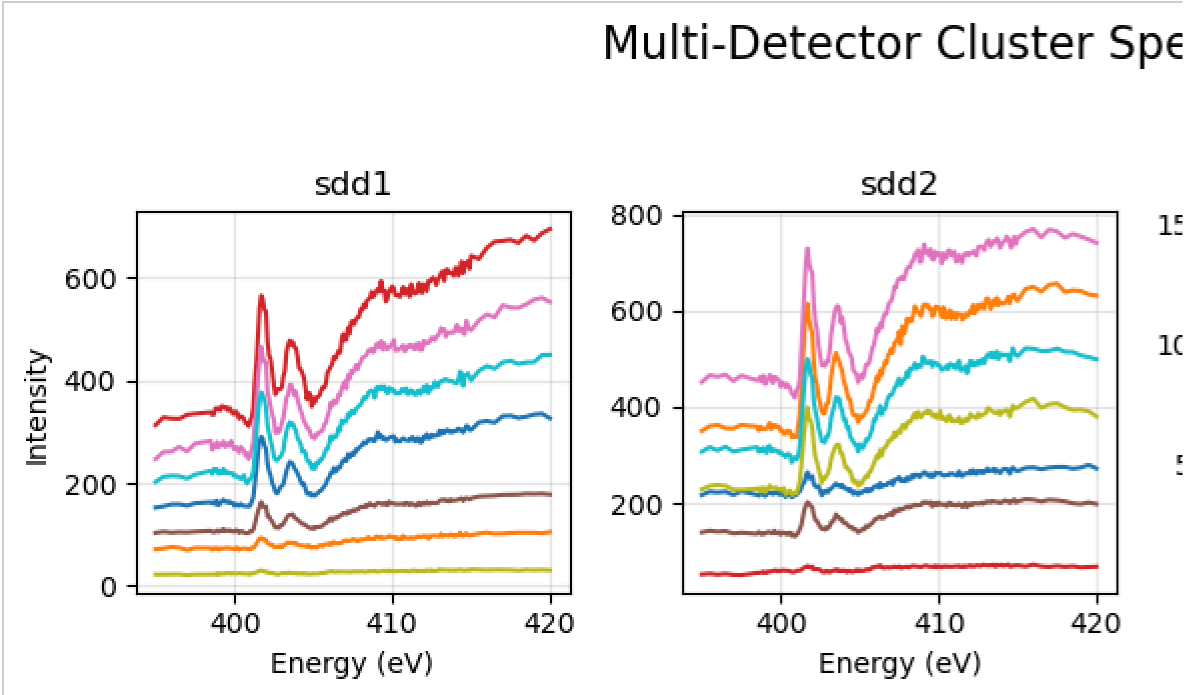



In [10]:
```
# For all detectors
cluster_results=run_clustering_all_detectors(pca_results, use_full_metadata=False,
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
============================================================
Running Multi-Detector Clustering Analysis
============================================================
Loading PCA results for 'sdd1' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd1 cluster results saved.
Loading PCA results for 'sdd2' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
    -> sdd2 cluster results saved.
Loading PCA results for 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> sdd3 cluster results saved.
Loading PCA results for 'sdd4' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
    -> sdd4 cluster results saved.
Loading PCA results for 'average' from D:/Data/Testing_jupyter-python-axis2000/N_sta
cks/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
    -> average cluster results saved.

Multi-detector clustering plots saved to:
  -> D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack\202
6-04-24_223037_stack_data_PyMca_cluster_comparison_maps.png
  -> D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack\202
6-04-24_223037_stack_data_PyMca_cluster_comparison_spectra.png
```



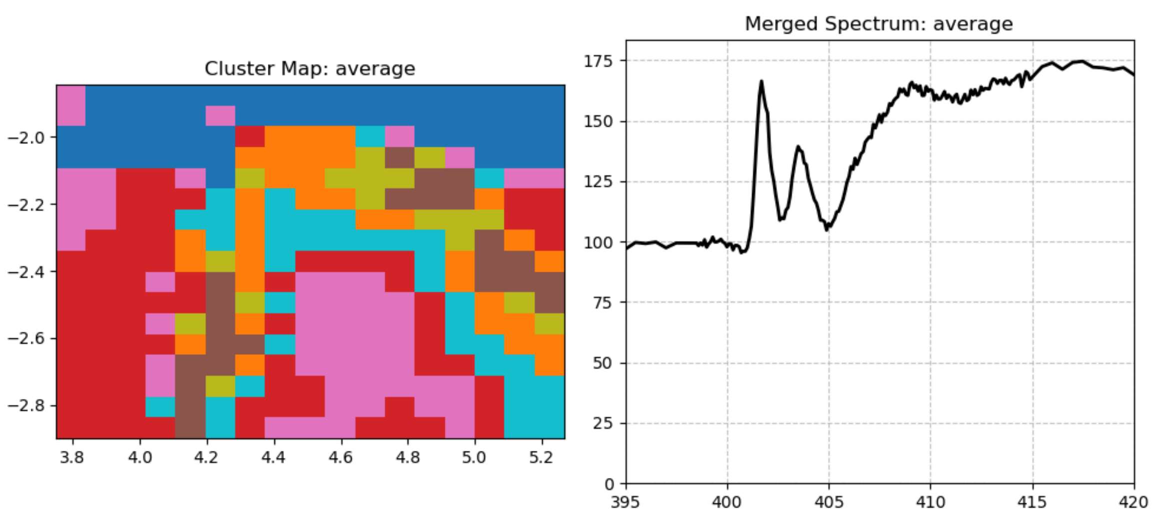



```
    -> Combined cluster sums saved to: D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_all_detectors_cluster_sums.csv
```

In [11]:
```
#For all detectors
interactive_cluster_merger(cluster_results)
```

```
VBox()
```

merger_fig_2726597772912




In [5]:
```
# Adds the I0 from the Au mesh (mcc1) from a number of  N1s spectra
# The reason is to increase the S/N ratio of the I0, thus increasing the S/N of the
sum_mcc1_spectra()
```

```
Loading 2026-04-24_223037_stack_data.h5...
Loading 2026-04-24_233217_stack_data.h5...
Loading 2026-04-24_205859_stack_data.h5...
Loading 2026-04-24_195642_stack_data.h5...
Loading 2026-04-24_182915_stack_data.h5...

[SAVE SUCCESS] MCC1 Average saved to:
--> D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\test
1-all.csv

[SAVE SUCCESS] MCC1 Average saved to:
--> D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\test
2-removed.csv
```

In [ ]:

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In [ ]: